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A Last-100-Meters User Acceptance Issue: Navigating Fully Autonomous Vehicles through on-Board Navigation Systems

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ABSTRACT

As the navigation interface of fully autonomous vehicles (FAV) will be wholly dependent on On-board Navigation Systems (ONS), it is foreseeable that ONS will replace steering wheels as the primary control mode of future FAVs. Thus, the ease of setting precise destinations using ONS may become a key factor affecting FAV user acceptance. A survey ($N=200$) comprising Navigation Anxiety Questionnaire and User Preference Questionnaire reveals the user's acceptable measurements of distance error and execution time when navigating FAVs in six common situations. Subsequently, an in-vehicle simulation test ($N=24$) found most participants in the simulation test performed worse than the user's acceptable measurements obtained through the questionnaire survey, which indicated that the existing ONS failed to meet users' expectations for the FAV navigation experience. Findings of the questionnaire survey and the simulation test reveal a severe "last-100 meters" issue when controlling FAVs by ONS, and that destination familiarity, gender difference, text input behavior, map scale, and map display mode should be seriously considered when designing ONS for FAV control.

KEYWORDS

Fully autonomous vehicle; on-board navigation system; last-100-meters; user acceptance; user experience

1. Introduction

The modern passenger vehicle, the most popular form of road transportation, was invented to help travelers reach their ideal destination quickly and easily (Happian-Smith, 2000). The fully autonomous vehicle (FAV) is now regarded as the future of the automotive industry by the public (Rajasekhar & Jaswal, 2015). Thanks to FAVs, people will obtain a better user experience by being released from the restriction of keeping their "eyes on the road, hands on the wheel" (González et al., 2007). Since human intervention will no longer be required, On-board Navigation Systems (ONS) are likely to replace steering wheels as the main control system for FAVs. The only thing people would need to do is to mark their destination precisely in the ONS before the driving begins (Qian et al., 2020). But there are two crucial issues that need to be solved before using ONS to effectively operate FAVs: (1) precise destination marking by the user; and (2) accurate self-localization and wayfinding by the FAV. Since precise localization and wayfinding are closely related to driving safety, a large amount of research has rightly focused on this field, and continuous progress has been made in the development of driving sensor technology (Campbell et al., 2018; Happian-Smith, 2000; Javanmardi et al., 2017, 2019; Kalra & Paddock, 2016). Unfortunately, current ONS have been designed mainly in reference to user interactions and behaviors in manual driving. Due to the huge difference between manual and fully

autonomous driving, how to set destinations accurately in the FAV's ONS is likely to become a key issue affecting user experience and acceptance (Qian et al., 2020).

1.1. Spatial knowledge and wayfinding strategy

The increasing, and potentially exclusive, use of ONS to navigate FAVs will place greater demand on people's spatial knowledge. Spatial knowledge is central to setting a destination, recognizing paths and landmarks, and being able to adjust routes in-the-moment while underway. Spatial knowledge is a vital ability for building up cognitive maps, which are the mental representations of the physical environment (Kitchin, 1994). It consists of three distinct elements: landmark, route, and survey knowledge (Hart & Moore, 1973; Siegel & White, 1975). Spatial knowledge is generally gained through observations collected as one travels through the environment. With spatial knowledge, people are able to find routes and relative positions, such as navigating from location A to B or guiding another person to a specific place (Kuipers, 1978; Wiener & Mallot, 2003). Spatial knowledge acquisition is an essential process in cognitive mapping construction, which helps people perceive, acquire knowledge and manipulate objects in space, and successfully transfer environmental and geographical data into their long-term memory (Downs & Stea, 2017; Wen et al., 2011).

Spatial cognitive ability is mainly reflected in map reading and wayfinding (Ben-Elia, 2021; Golledge, 2003; Saucier

et al., 2003). Psychological studies reveal that spatial cognition: (1) varies among individuals; and (2) could cause obvious differences in navigation results (ADAM, 1999; Carroll, 1993; Dabbs et al., 1998; Lloyd & Bunch, 2008). Therefore, Lawton et al. designed the Navigation Strategy Questionnaire and Navigation Anxiety Questionnaire to evaluate map reading and wayfinding strategy abilities, and found gender differences in wayfinding (Lawton, 1994, 1996; Lawton & Kallai, 2002). Several researchers also found that male and female participants differ in their wayfinding performance when facing familiar versus unfamiliar locations (Chang, 2013; Matthews, 1987).

1.2. Map style and effectiveness

Prior research results indicate that map style has an impact on navigation results (Montello, 2002). First, low map precision results in users' inability to accurately mark target locations (Devlin & Bernstein, 1997; Greenfeld, n.d.; Shen & Stopher, 2014). Moreover, Padilla et al. found that differences in map scale had an impact on map reading performance, with women more easily affected by scale than men (Padilla et al., 2017). Second, inaccurate map data and unmarked route changes could be sources of travel problems (Bonin-Font et al., 2008; Rossano & Warren, 1989; Warren & Scott, 1993). Third, the map interface (including color, text, layout, etc.) is a key determinant of user experience. Thus, Shu et al. indicated that the "last-mile" navigation problem already plagued people due to the lack of micro-map information (Shu et al., 2015).

Several researchers have proposed strategies and guidelines to improve map readability (Jenny et al., 2008; Krygier & Wood, 2016; Peterson, 2020). Jenny et al. proposed 10 design principles for internet maps, with guidance for readability through the choice of dimensions and distances, and for generalizability through shape simplification and information density, among other considerations (Jenny et al., 2008). Garland et al. examined the effectiveness of color and street detail on a city bus map, finding tradeoffs in trip planning accuracy, difficulty, frustration, and confidence (Garland et al., 1979). Kulhavy et al. found that textual information can be used to prompt retrieval of relevant information, making it easier to recall the location-related representations in working memory (Kulhavy et al., 1993). More recently, Lin et al. indicated that map interfaces for ONS had a significant impact on navigation results and user satisfaction, with again, the operation performance and user satisfaction of male participants were significantly higher than those of female participants (Lin & Chen, 2013; Lin & Chien, 2010).

Autonomous driving systems are striving to leverage millimeter-level high-definition maps (Seif & Hu, 2016). But the ONS/CGM (client-side GPS map) features that this level of detail affords are not yet applicable to meter-level (or higher) navigation and wayfinding tasks in autonomous driving. While prior literature suggests it is convenient to control FAVs by ONS, the inability to precisely locate true destinations and make minor route adjustments are

therefore likely to cause a "last-100-meters" issue. Navigation distance error and execution time are generally tested as the key indicators of user experience in navigation or pathfinding tasks (Atoum, 2023; Ishikawa et al., 2008). But it is unknown how differences between users' tolerable limits for these indicators and their practical performance on them will affect experience and anxiety. Therefore, the specific research questions of this study are:

1. Will user performance (including navigation distance errors and execution time) in the simulation test exceed the user's acceptable measurements obtained by questionnaires when navigating FAVs via ONS?
2. What factors might affect user navigation responses and operating performance when control FAVs, resulting in the "last-100-meters" issue?

2. Method

To address these questions, we developed a study comprising a questionnaire survey and an in-car simulator test. In the questionnaire survey, we first investigated users' acceptable measurements of distance error and execution time in six common situations, including new or unfamiliar address, severe weather, carrying bulky or heavy items, poor physical or mental state, pick up/drop off the elderly or disabled, and in a hurry (Park & Kim, 2014). Among the six situations, five were not suitable to be tested as independent variables in the simulation test because they are difficult to detect (carrying bulky or heavy items, poor physical or mental state, and in a hurry) or cannot be implemented (severe weather, pick up/drop off the elderly or disabled). For instance, people have different standards for heavy objects due to their own strength differences, resulting in an inability to choose a suitable heavy item for testing. Alternatively, since all the participants are from the same campus, address familiarity can be accurately distinguished through a pre-questionnaire. In addition, address familiarity is the most frequently encountered situation among the six situations, so the results will be more valuable for reference. Therefore, we tested address familiarity as the independent variable in the simulation test to obtain participants' operating performances, and to determine whether their performance exceeded the acceptable measurements they indicated in the questionnaire survey.

2.1. Questionnaire survey

2.1.1. Participants

A total of 213 questionnaire invitations were randomly distributed on the sidewalk of a large university campus in China, with 200 qualified questionnaires (81 males and 119 females) collected. Invited participants would scan a QR code to open an online questionnaire¹ which logged their responses, timing, user ID, and IP address. All participants were informed that their private information (user ID and IP address) would be partially displayed to researchers but used only to ensure the uniqueness of respondents. The 13



Figure 1. An 11-inch iPad Pro setup in an autonomous driving simulator.

disqualified participants claimed that they never had a driver's license yet gave conflicting answers to subsequent questions about their driving frequency. The mean age of qualified participants was 22.8 ($SD = 2.4$), 68 of whom had never obtained a driving license.

2.1.2. Apparatus

Two questionnaires developed by UX specialists and psychologists, including the Navigation Anxiety Questionnaire and User Preference Questionnaire, were adopted to measure participants' acceptable values for variables that influence experience when using ONS to control FAVs (Ishikawa, 2019; Lawton, 1994).

(1) Navigation Anxiety Questionnaire

The Navigation Anxiety Questionnaire consists of 6 common situations that might cause user anxiety when more distance errors than acceptable limits occur, such as when setting a destination in a new and unfamiliar area. The scale for anxiety level is "1 = not at all anxious, slightly anxious, moderately anxious, somewhat anxious, 5 = very anxious".

(2) User Preference Questionnaire

The User Preference Questionnaire consists of 4 items: acceptable distance error, acceptable execution time, expected map scale and expected control mode. Regarding the options for execution time, a pre-questionnaire survey distributed to the different participants ($n = 20$) found a distance error of 50 meter and an execution time of 60 s as the maximum acceptable value. Thus, the choice of acceptable distance error is "1 = less than 1 meter, 2 = 1–10 meter, 3 = 11–20 meter, 4 = 21–50 meter, 5 = more than 50 meter" and "1 = less than 10 s, 2 = 11–20 s, 3 = 21–30 s, 4 = 31–40 s, 5 = 41–50 s, 6 = 51–60 s, 7 = more than 60 s".

2.1.3. Procedure

All participants were required to independently complete the Navigation Anxiety Questionnaire and User Preference Questionnaire in sequence online. After completing the two questionnaires, each participant completed a brief personal information survey, including gender, age, possession of a driver's license, and frequency of driving.

2.2. In-car simulator test

2.2.1. Participants

A total of 24 participants (12 males and 12 females) were recruited online through the college's student online chat group or offline through an on-campus recruitment poster. The mean age of the qualified participants was 23.2 ($SD = 3.3$). All participants were identified as local and proficient in ONS/CGM operation by a brief preliminary survey. The preliminary survey comprised several single-choice questions eliciting participants' frequency of use of ONS/CGM and familiarity with several buildings pre-selected by the research team.

2.2.2. Apparatus

An 11-inch iPad Pro (Wi-Fi + Cellular, built-in GPS) was used as an ONS to navigate a simulated FAV within a real Audi A6 (See Figure 1). The vehicle was parked in an outdoor parking lot throughout the test to ensure unobstructed reception of cellular data and GPS. The Apple iPad's built-in map, whose interface and map scale (1:500) are consistent with popular ONS/CGS, was used as the experimental map. A digital camera and built-in screen recording application captured participants' interactive activities and execution time without interruption. As with the questionnaire survey, all of the materials in the simulator test were in the native language of the participants (Chinese).

2.2.3. Procedure

Before the test officially started, participants were orally informed of the experiment's instructions and given about 10 minutes to familiarize themselves with the simulator, apparatus, and software. Participants were asked to complete the assigned navigation tasks (1 and 2) in the shortest time with the minimum distance error. The navigation destination in task 1 was a familiar building and in task 2 was an unfamiliar building. The familiarity of each building had been confirmed through the pre-questionnaire survey. Both were relatively common buildings in the local area and were located 5 km away from the simulator. Each task provided participants with a task guide manual containing a detailed textual description (including gate name/number, building

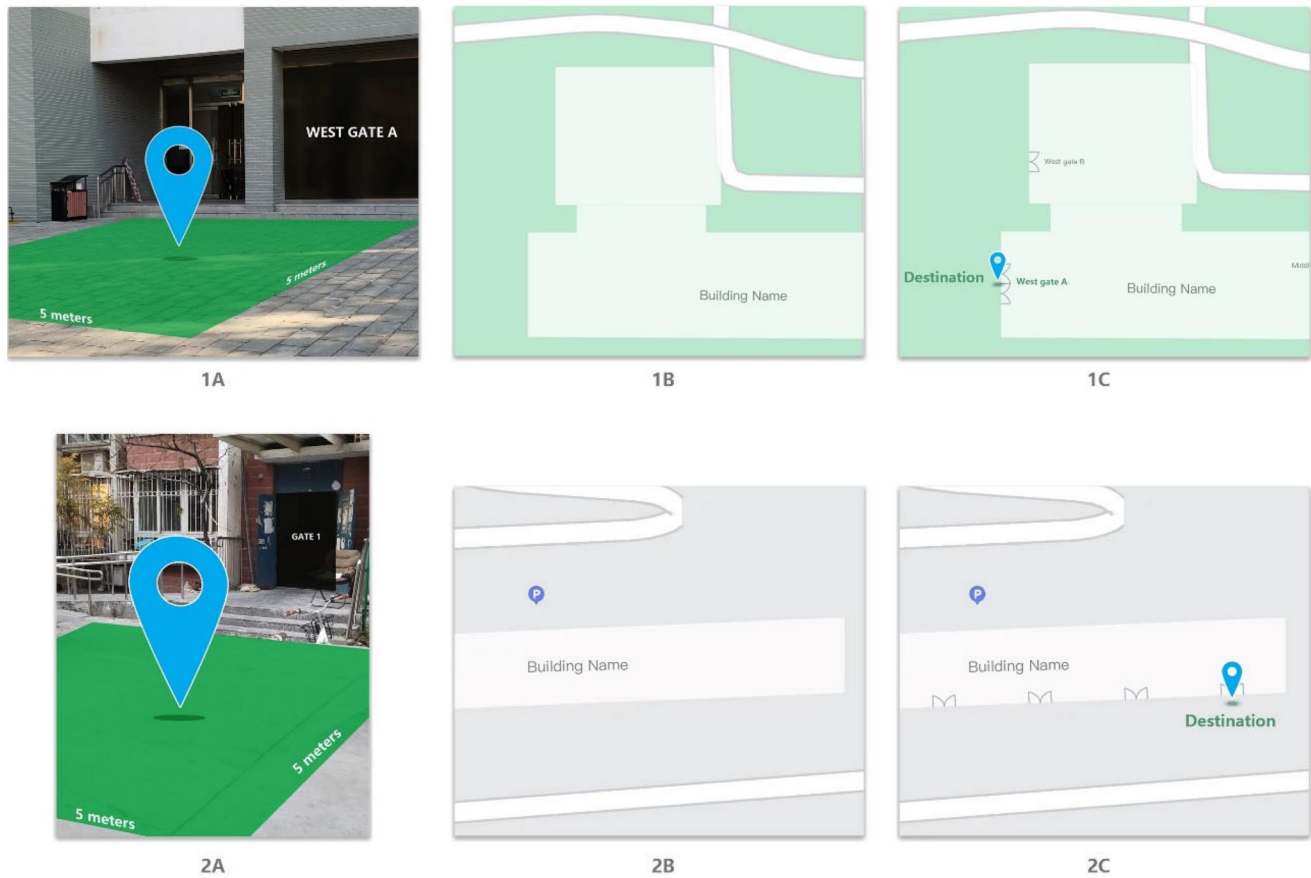


Figure 2. Schematic diagrams (partial). 1A and 2A: Photos provided to participants in task 1 and 2 guides. 1B and 2B: 2D map view of the destination buildings displayed in the ONS (1:500). 1C and 2C: 2D map view of the destination buildings revised by the research team to include the missing destination entrances.

name/number, community name/number, street name/number, district name/number, and city name) and a photo of the destination (1A or 2A shown in Figure 2). 1B and 2B in Figure 2 showed the assigned destinations displayed in the ONS (iPad) at the maximum scale of 1/500. Since all the gates of the selected buildings (1C or 2C shown in Figure 2) were not marked in the ONS (iPad), participants had to set the destinations precisely in the ONS (iPad) relying only on experience, the provided pictures (1A or 2A), text descriptions, and the map itself (1B or 2B). To avoid distance errors caused by the GPS signal, ONS (eg, route automatic correction), and other factors, the research staff used map ranging to measure distance errors between the destinations set by the participants and the assigned destination (distance error = measured distance $\times$ map scale) (Canters, 2002).

3. Results

3.1. Questionnaire survey

We analyzed gender differences in the Navigation Anxiety Questionnaire by *t*-test (Lawton & Kallai, 2002), and the User Preference Questionnaire by Fisher's exact test (Ek, 2015). All data were analyzed using SPSS (95% confidence interval) (Field, 2013).

3.1.1. Navigation Anxiety Questionnaire

Table 1 shows the *t*-test analysis of anxiety levels for male and female participants when distance errors exceeding acceptable values occurred in six situations. Female participants' anxiety appears to more readily increase when distance errors exceed their maximum acceptable values. In addition, significant differences were found between male and female participants in all situations except for the situation of severe weather.

Statistical results show that all mean anxiety scores by participants were higher than 4 points (on a 5-point scale). The situation of poor body/mental state was rated as the cause of the lowest anxiety level for both female and male participants. In contrast, carrying bulky or heavy items was the cause of the highest anxiety level for female participants, whereas being in a hurry was the most concerning situation for male participants.

3.1.2. User Preference Questionnaire

3.1.2.1. Acceptable distance error. No significant variables were found by gender in six common navigation scenarios (see Table 2). The proportion of male participants choosing the option with an error of less than 20 m ranged from 60.5% (poor physical or mental state) to 81.5% (carrying bulky or heavy items). The proportion for female

Table 1. Descriptive statistics for Navigation Anxiety Questionnaire.

Question	Situation	Gender	N	x	SD	t	p
What will your anxiety level be if the parking distance error exceeds your maximum acceptable value?	New or unfamiliar address	Male	81	4.15	0.94	-2.79	0.006*
		Female	119	4.47	0.70		
	Severe weather (eg, rainy or snowy)	Male	81	4.31	0.96	-1.07	0.29
		Female	119	4.44	0.74		
	Carrying bulky or heavy items	Male	81	4.32	0.82	-3.54	0.001*
		Female	119	4.67	0.58		
	Poor physical or mental state	Male	81	4.07	0.92	-2.95	0.004*
		Female	119	4.44	0.81		
	Pick up/drop off the elderly or disabled	Male	81	4.30	0.91	-2.19	0.03*
		Female	119	4.55	0.74		
	In a hurry	Male	81	4.35	0.88	-2.79	0.006*
		Female	119	4.65	0.65		

Table 2. Descriptive statistics for acceptable distance errors in six situations.

Question	Situation	Gender	N	df	Value	p
What is the maximum distance error you can accept between the destination set in the ONS/CGM and the actual destination you want?	New or unfamiliar address	Male	81	4	3.43	0.50
		Female	119			
	Severe weather (eg, rainy or snowy)	Male	81	4	0.70	0.96
		Female	119			
	Carrying bulky or heavy items	Male	81	4	3.49	0.48
		Female	119			
	Poor physical or mental state	Male	81	4	4.59	0.33
		Female	119			
	Pick up/drop off the elderly or disabled	Male	81	4	0.64	0.97
		Female	119			
	In a hurry	Male	81	4	1.92	0.76
		Female	119			

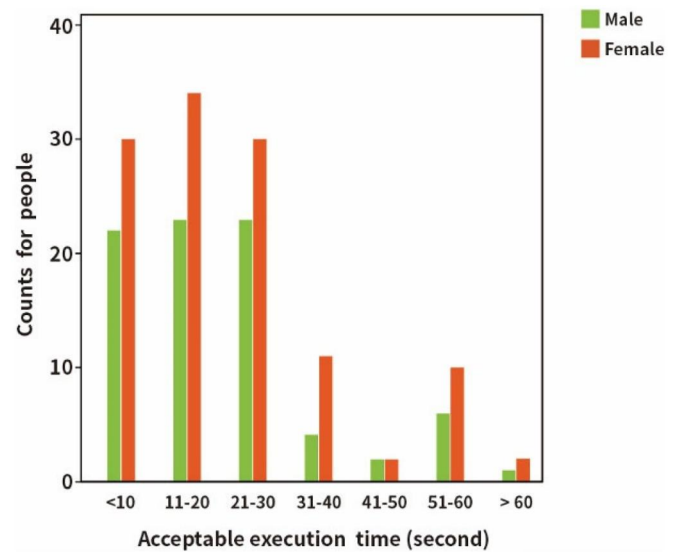
participants ranged from 71.4% (severe weather) to 80.7% (pick up/drop off the elderly or disabled). A higher proportion of female participants appear to prefer smaller distance error than male participants, except for the situation of carrying bulky or heavy items. Overall, statistical results show that a distance error of more than 20 meters was unacceptable for most users in general.

3.1.2.2. Acceptable execution time. Question 2 of the User Preference Questionnaire reveals how long people are willing to spend on overall operation of the ONS. In total, 81% of participants, including 84% of male participants and 79% of female participants, refused to spend more than 30 s on the ONS operation. There was no significant difference in execution time between male and female participants, $X^2(6, N=200) = 1.95, p = 0.95$.

Male participants appeared to be more critical of exceeding acceptable execution time than female participants, as 66.7% of male participants and 53.1% of female participants believed that the execution time of the ONS should be no more than 20 s, while only 7.4% of male participants and 16.3% of female participants would spend over 40 s on the operation of the ONS (see Figure 3).

3.1.2.3. Expected map scale. As shown in Figure 4, only 6% of participants believed that the current map scale of 1:500 was sufficient for operating an FAV, and no significant difference in map scale was found by gender, $X^2(6, N=200) = 6.91, p = 0.31$.

In this category, more male participants expected a higher map scale for the ONS than female participants, with

**Figure 3.** Acceptable execution time on the User Preference Questionnaire.

95.1% of male participants and 93.3% of female participants preferring an ONS with a scale of 1:100 or higher when searching for a parking spot.

3.1.2.4. Extra control mode. Overall, 87.5% of participants agreed that an FAV cannot be controlled by the ONS alone. Fisher's exact test shows there was a statistically significant difference among male and female participants, $X^2(3, N=200) = 9.29, p = 0.019$ (see Figure 5).

As an intuitive control mode, results show that the ONS was more popular among male participants than female

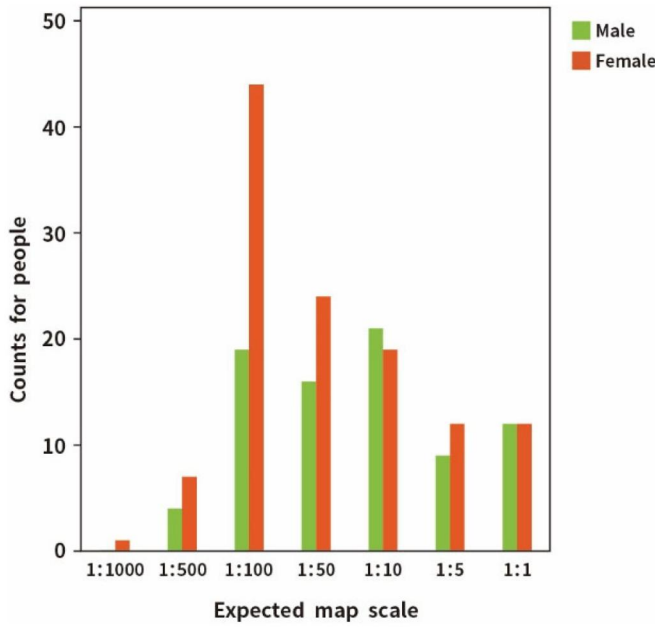


Figure 4. Expected map scale by gender on the User Preference Questionnaire.

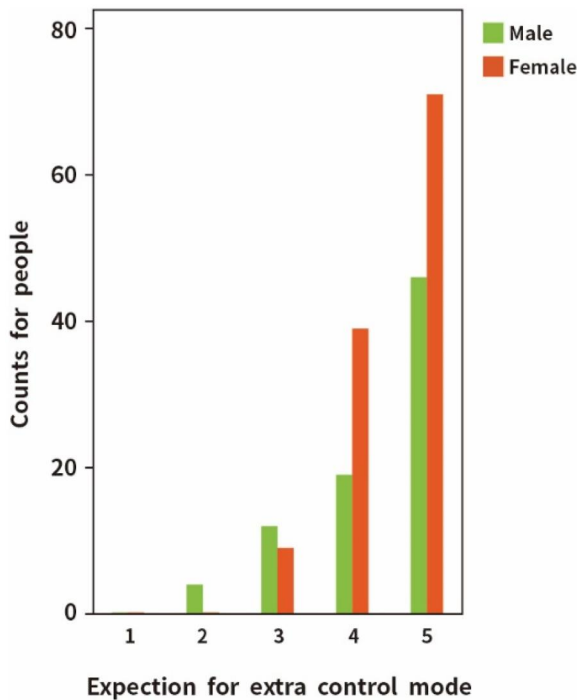


Figure 5. Expectation for extra control modes by gender on the User Preference Questionnaire. 1 = definitely not, 2 = probably not, 3 = might or might not, 4 = probably yes, 5 = definitely yes.

participants. However, 80.2% of male participants and 92.4% of female participants believed that extra control modes should be provided in FAVs.

3.2. In-car simulator test

We analyzed the significant differences of distance error (meter), execution time (second), and user evaluation by *t*-test at the 95% confidence interval. All data were analyzed using SPSS.

Table 3. Descriptive statistics for distance error in the simulation test.

Task	Gender	N	X	SD	t	p
Task 1 (A familiar building)	male	12	4.4	4.8	-3.42	0.002*
	female	12	14.3	8.7		
Task 2 (An unfamiliar building)	male	12	43.5	24.5	-1.66	0.11
	female	12	58.4	19.3		

3.2.1. Distance error

54.2% participants (13 of 24) used a dual-maps navigation strategy: they first found the target buildings' approximate locations using the Standard map, then slightly adjusted to find the precise destinations using the Satellite map. The other 11 participants completed the tasks using only the Standard map.

One outlier caused by a female on task 1 was found by transforming scores into *z*-scores. *z*-scores > 3.0 were classed as outliers and were replaced by mean score in the data set (Tabachnick et al., 2007).

As displayed in Table 3, in task 1, male participants ($M=4.4$, $SD=4.8$) had significantly better performance than female participants ($M=14.3$, $SD=8.7$), $t(22) = -3.42$, $p=0.002$. In task 2, no significant difference was found between male and female participants, $t(22) = -1.66$, $p=0.11$. The mean distance error of male participants was 43.5 ($SD=24.5$), somewhat smaller than that of female participants ($M=58.4$, $SD=19.3$).

Generally, results of the paired *t*-test found that notable differences appeared between task 1 ($M=9.3$, $SD=8.5$) and task 2 ($M=50.9$, $SD=22.9$), $t(23) = -8.48$, $p<0.0001$. In particular, regardless of whether participants were male ($t(11) = -5.06$, $p<0.0001$) or female ($t(11) = -7.00$, $p<0.0001$), the distance errors in task 1 were significantly smaller than those in task 2.

3.2.2. Execution time

Most participants completed the navigation tasks by entering text (the building title) first, and then moving the cursor to more precisely locate the destination. Therefore, the analysis of execution time was divided into two parts: text input time and cursor movement time.

One outlier caused by a female on task 1 (text entry), two outliers by a male on task 2 (for text entry and cursor movement), and two outliers by a female on task 2 (for text entry and cursor movement) were found by transforming scores into *z*-scores. *z*-scores > 3.0 were classed as outliers and were replaced by mean score in the data set (Tabachnick et al., 2007).

As shown in Table 4, execution time, including the duration of text typing and cursor moving, did not differ significantly between male and female participants in both tasks.

Notable differences in execution time between task 1 ($M=57.0$, $SD=26.2$) and task 2 ($M=106.9$, $SD=21.0$) were founded by the paired *t*-test, $t(23) = -6.85$, $p<0.0001$. Regardless of whether a participant was male ($t(11) = -5.27$, $p<0.001$) or female ($t(11) = -4.29$, $p=0.001$), execution time in task 1 was significantly shorter than in task

Table 4. Descriptive statistics for execution time in the simulation test.

Task	Process	Gender	N	x	SD	t	p
Task 1 (A familiar building)	text entry	Male	12	12.3	10.5	-0.55	0.59
		Female	12	14.8	11.7		
	cursor movement	Male	12	44.0	20.9	0.11	0.92
		Female	12	43.0	24.6		
	execution time	Male	12	56.3	23.2	-0.14	0.89
		Female	12	57.8	30.0		
Task 2 (An unfamiliar building)	text entry	Male	12	52.2	14.1	-1.16	0.26
		Female	12	57.8	9.3		
	cursor movement	Male	12	56.6	20.2	1.21	0.24
		Female	12	47.3	17.6		
	execution time	Male	12	108.8	22.7	0.42	0.68
		Female	12	105.1	20.0		

Table 5. Descriptive statistics for user evaluation in the simulation test.

Task	Gender	N	x	SD	t	p
Task 1 (A familiar building)	male	12	3.25	0.62	-1.27	0.22
	female	12	3.58	0.67		
Task 2 (An unfamiliar building)	male	12	2.50	0.91	-0.23	0.82
	Female	12	2.58	0.90		

2. Specifically, male participants spent significantly less time on text entry in task 1 than they did in task 2, $t(11) = -12.05$, $p < 0.0001$, as did female participants, $t(11) = -9.00$, $p < 0.0001$. Conversely, no obvious differences were found in cursor movement time between task 1 and task 2 for male and female participants.

3.2.3 User evaluation

Regardless of which task was performed, there were no significant differences in user evaluation for destination searching by gender (See Table 5). Male participants evaluated slightly lower than female participants in both tasks.

Overall, there were significant differences between task 1 and task 2, $t(23) = 4.32$, $p < 0.0001$, with male participants ($t(11) = 2.28$, $p < 0.05$) and female participants ($t(11) = 4.06$, $p < 0.005$) both scoring much lower in task 2 than task 1.

4. Discussion

4.1. Is the “last-100-meters” a real issue?

Although the questionnaire survey results confirm that “50 meters” is the maximum acceptable value for distance error among users, the inevitable “last-100-meters” operating distance error is the key cause affecting user acceptance when navigating FAVs through ONS. As a consequence, the “last-100-meters” will become a real problem for navigation in future FAVs. In addition, the “last-100-meters” issue will be reflected not only in the distance error exceeding the maximum value deemed acceptable by users, but also in the high cost of interaction time and poor user interaction experience, expressed through anxiety level.

In our study, we conclude that five factors, including destination familiarity, gender, text input behavior, map scale, and map display mode, could cause the “last-100-meters” issue when navigating FAVs via ONS.

4.1.1. Destination familiarity

Destination familiarity is an easily underestimated or even ignored factor in fully autonomous driving scenarios. In the questionnaire survey, statistical results showed that the maximum distance error generally accepted by users was less than 20 meters. To be consistent with existing research (O'Neill, 1992; Prestopnik & Roskos-Ewoldsen, 2000), results of the in-vehicle simulation test showed that most participants successfully reduced distance errors to under 20 meters in the familiar destination task, but distance errors were significantly more than 20 meters when they faced an unfamiliar destination. However, even when faced with a very familiar destination, participants were challenged to avoid accidental distance errors caused by false memory or map reading errors. For instance, in the simulation test, one female participant had a distance error of 109.4 meters. The probability of similar errors in real scenarios is not high, but once they occur, they may cause unnecessary troubles for users. Furthermore, the results of the in-vehicle simulation test showed that whether it was a familiar or unfamiliar destination, the execution time was much longer than the 30 seconds accepted by participants in the questionnaire survey. We also found that when faced with a familiar task, participants spent a lot of time on cursor movement to find a suitable parking spot. We have reason to believe that in the future, with the increasing proportion of ONS and the continuous enrichment of map details, the time for cursor movement will be greatly reduced, thus lowering overall execution time to under 30 s, which was the key acceptable metric for execution time. But when faced with an unfamiliar destination, the time spent on text entry by participants was much more than this, so even if performance improvements in ONS can reduce cursor movement time, overall execution time for unfamiliar destinations is unlikely to be reduced to less than 30 s.

4.1.2. Gender difference

Gender differences were exposed by both the questionnaire survey and the simulation test. Although the questionnaire survey results showed that there was no significant difference in the acceptance of distance error between male and female users, female users were more likely to experience anxiety when the distance errors exceeded their acceptable limits (Padilla et al., 2017). In the simulation test, the familiar address task aimed to test users' spatial ability and map

reading ability (Lloyd & Bunch, 2008), while the unfamiliar address task reflected users' map-based reasoning ability and map reading ability (Ishikawa, 2016; Matthews, 1987). The performance of male participants was generally better than that of female participants in the task for a familiar address: first, the mean distance errors of male participants were significantly smaller than those of female participants; second, male participants commonly spent less time completing the task than female participants, but the difference was not significant. In contrast, in the unfamiliar address task, the differences in distance error and execution time between genders were not obvious. In detail, male participants were slightly better than female participants in distance error, while female participants took slightly less time than male participants. The performance of both male and female participants fell below the results from the questionnaire survey, which was significantly worse than that of the familiar address task.

4.1.3. Text input behavior

The results of the questionnaire survey and the simulation test showed that the combination of text input first and cursor movement later was the most common interaction pattern for map navigation. It was found that text input behavior was a vital cause for the large difference in execution time for the different tasks. Statistical results show that the time to enter text for the familiar address task was significantly lower than for the unfamiliar address task for both male and female participants (Lopez et al., 2021). In contrast, there was no significant change in the time to move the cursor between tasks. As shown in Figure 6, for the familiar address task, most participants quickly entered

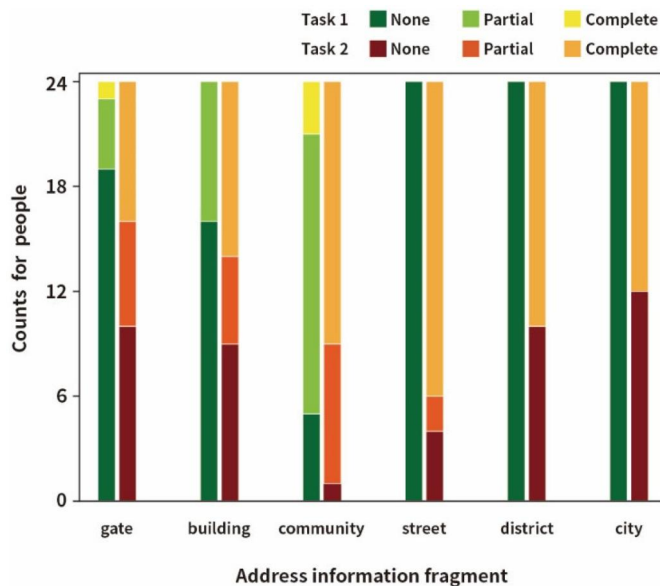


Figure 6. Statistics for address information input by task. The fragments of address information for both tasks consisted of gate name/number, building name/number, community name/number, street name/number, district name/number, and city name. Participants who did not enter any address information were marked as "none", those who entered partial information but incomplete were marked as "partial", and those who entered full address information were marked as "complete".

keywords of the target building and community, ignoring street, district, and city, thus greatly reducing the time spent on text input. 16.7% of participants did not even enter text information, but directly moved the cursor to find the target building on the map. Since participants had already established a cognitive map of the familiar address in their long-term memory, they could locate the building quickly and accurately by comparing the reference displayed on the map without a text description. In the unfamiliar address task, participants were more cautious in entering the address information to ensure an accurate location. No less than 50% of participants entered the city name and district name completely, whereas none did the same for the familiar address task. For the unfamiliar address, participants could not extract the target-related information from their memory to match with the map. Thus, they were constrained to rely on exogenous information (eg, text descriptions or pictures) when reading the digital maps.

4.1.4. Map scale

Map scale is a key factor impacting user performance and experience when controlling FAVs by ONS. Although our study did not conduct a test of user performance at different scales, results indicated that the current map scale (1:500) adopted by most ONS and mobile maps was unable to display sufficient detail and cannot meet the qualification for FAV navigation. The user questionnaire survey results showed that most participants (94%) believed the map scale should not be lower than 1:100; in the simulation test, over 80% of participants considered the 1:500 map scale an inconvenience for accurate location. Existing research points out that people need some references in wayfinding, whether in a natural or virtual environment (Dabbs et al., 1998; Lawton, 1996). Some studies also confirm that map scale differences affect map reading performance, with females more likely to be influenced by map scale than males (Padilla et al., 2017; Saucier et al., 2003). Insufficient map scale causes an incomplete display of geographic information, thus affecting the participants' map reading performance. As shown in Figure 7, in the simulation test, the gates of the target building of the two tasks were not marked on the ONS, which hindered participants in completing the tasks. In the familiar address task, participants were able to find the location by relying on their memory. In the unfamiliar address task, without any spatial memory related to the target building, participants were forced to speculate or accept the system's default settings, and which resulted in uncontrollable distance errors.

4.1.5. Map display mode

The study also found that different map display modes have an impact on user positioning errors. At present, ONS generally provides Standard and Satellite map modes. In Standard map mode, buildings, vegetation, roads, etc. are composed of lines and colored blocks, which makes it difficult for participants to distinguish a building's characteristics such as color, form, and gates. Furthermore, a top-down

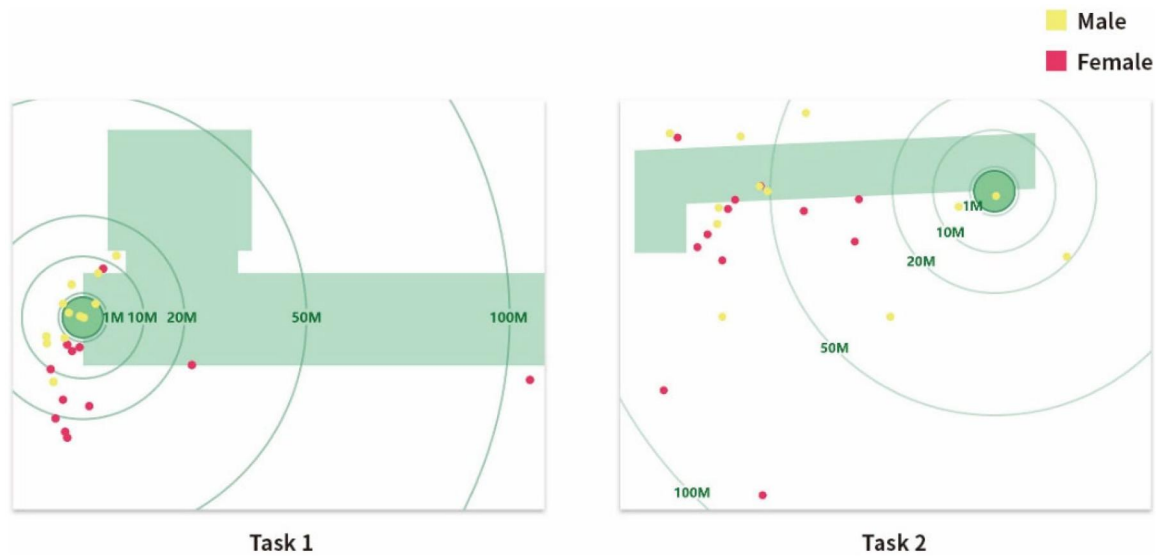


Figure 7. Parking spots marked by participants in the simulation test.

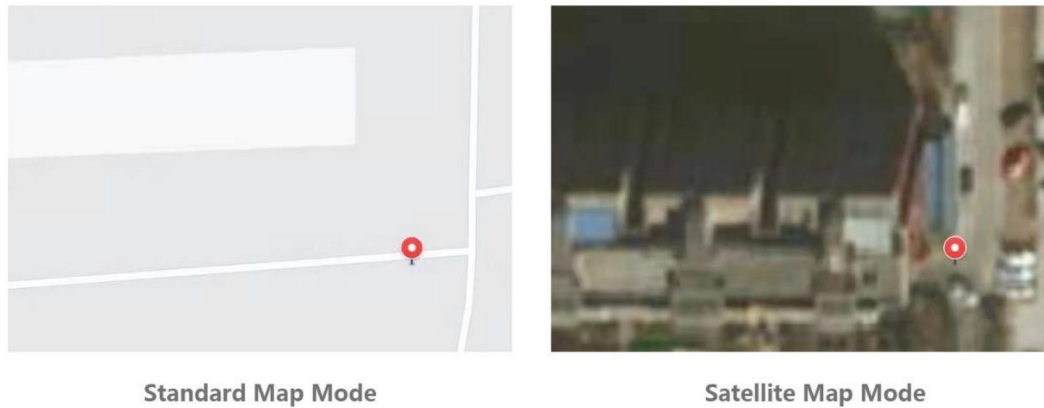


Figure 8. The different display modes at the maximum scale of 1:500 (partial).

view is quite different from a driver's (egocentric) perspective of the real world, which increases reading difficulty for participants. In Satellite map mode, the provided satellite photos can show a building's real appearance, which partially helps users to accurately set the destination. In the simulation test, 54.2% of participants searched for the task-assigned gates in Satellite Map Mode. However, there are also several drawbacks in Satellite map mode (as shown in Figure 8), which our test observed: (1) The resolution of the satellite images was insufficient for precise positioning, relative to Standard maps. The 1:500 scale satellite image was so blurry that participants could barely distinguish architectural characteristics; (2) Satellite images often include perspective distortion (especially for high-rise buildings), which caused a deviation in participants' precise positioning; (3) Satellite images provided by ONS normally have a monthly-update lag, which may cause a notable difference between the satellite images and the actual environment, thus affecting users' pathfinding performance. For example, a building under construction may change its appearance within a short period of time. The buildings displayed on the satellite images could be quite different from the actual buildings

because of the update lag, which may cause obstacles for pathfinding.

4.2. Is ONS the best control mode for FAVs?

The latent distance errors and user experience issues in FAVs raise a serious concern: should control by ONS be considered the "gold standard," or the exclusive control mode, for FAVs in the steering-free future? While the ONS provides great convenience, it also brings many user experience issues to everyone: in addition to the "last-100-meters" issue, high learning costs, unintuitive and unfriendly user interfaces, user privacy disclosure concerns, and other issues may cast a shadow over the popularity of FAVs.

At present, both automobile navigation systems and mobile maps are designed for manual driving, walking, or cycling. Travelers can make short-distance adjustments easily and accurately without maps. For example, when driving manually, users can control vehicles through the steering wheel; while walking or cycling, users can also adjust their location simply and quickly. However, for FAVs, when users do not have a driving license or are not skilled at

driving—both of which are likely to become more prevalent as vehicles assume a larger role in driving—adjusting the location of the vehicle will become a new pain point of user experience in new contexts. This research shows that navigation by ONS could be a good option when commuting between familiar destinations, such as home and work. However, for a new or unfamiliar destination, driving by ONS is likely not to be a user-friendly solution to most users. For instance, it is difficult for users to precisely locate destinations within open spaces that lack conspicuous landmarks (eg, rural suburbs, large squares, etc.). Meanwhile, precise localization forces users to obtain a mass of spatial information for assistance before driving, which will significantly increase communication costs and cognitive loads, especially for people with poor spatial perception ability. In addition, detailed data collected by cameras and other driving sensors raise public privacy and security concerns (Lim & Taeihagh, 2018). It is likely that governments will introduce policies to restrict data collection, which may cause some data (especially the kind of data displayed in an ONS) to become unavailable. Furthermore, uncertain and dynamic navigation needs sometimes have an impact on navigation results (Kiefer et al., 2017). In manual driving, people have formed an ordinary and convenient method of “map and steering wheel” to address the “last-100-meters” issue: they generally set a rough destination by ONS before driving, and then control the vehicle by manipulating the steering wheel personally, or by guiding drivers using voice and bodily gestures instead of the ONS, when approaching the intended destination (Qian et al., 2020). But due to the high operational cost and poor experience of entering or changing destinations in GPS maps, people generally do not make short-distance corrections in the ONS once underway (Colley et al., 2022; Terada et al., 2006).

Research on design thinking indicates that empathy is the first step in creating high-quality user experiences (Brown, 2008; Lewrick et al., 2018). Knowing that drivers will be gradually replaced by in-vehicle AI with the advancement of autonomous driving technology, instead of forcing people to navigate FAVs via ONS only, we strongly recommend that automotive enterprises introduce voice or gestures as an additional control mode for FAVs. How to make user interaction for in-vehicle AI follow the user behaviors established in manual driving, and provide a better user experience than manual driving should become one of the key factors for FAVs to be accepted by the public (Zhang et al., 2022). In the future, along with good UX, FAVs should provide multiple modes of interaction, to account for drivers' different capabilities and preferences (Colley et al., 2022; Detjen et al., 2021).

Through a literature analysis, questionnaire survey, and simulator test, we found that a variety of factors, including user spatial cognition, interaction mode, map scale, and GPS accuracy, contribute to causing a “last-100-meters” issue in the forthcoming era of the FAV. Some factors, like map scale and GPS accuracy, will be solved by the iteration of software and technology. However, user-related factors, such

as spatial cognition and interaction mode, cannot be solved perfectly just by iterating on current technological solutions.

5. Conclusion

With the continuous advancement of technology, FAVs seem likely to become the dominant mode of future on-road travel. Through literature analysis, questionnaire survey and user simulation test, we proposed a research question that a potential “last-100-meters” issue exists in ONS navigation of FAVs, and verified its existence. Additionally, we found destination familiarity, gender difference, text input behavior, map scale, and map display mode had an impact on user performance and user experience when navigating FAVs through ONS.

Although we exposed how user experience issues can influence the acceptance of FAV, the study still has limitations. First, the participants involved in our study cannot fully represent the diversity of possible driver types or personas. All of our participants were between the ages of 20-30 years and were experienced in ONS navigation. It is possible that elderly and disabled drivers may become a source of user growth in the coming years. Therefore, user personas including age, culture, education, and other personal characteristics need to be explored through further studies. Second, the scenarios in the simulated test cannot fully represent all driving, navigating and geographic scenarios encountered in real life. Future work should thus focus on enriching the breadth of usage scenarios to extend the scope of our research. In addition, we hope others will explore more factors that may contribute to the “last-100-meters” issue. Recently, some scholars have observed that when faced with unfamiliar addresses, the performance of elderly people in road recognition drops significantly (Lopez et al., 2021). Although autonomous driving system can make up for a lack of driving ability, the low-user-experience of Standard maps will become an obstacle to their control of FAVs. This challenge could reduce the attractiveness of FAVs to the elderly, who otherwise would be among their main beneficiaries and consumers (Kovacs et al., 2020). Besides the elderly, physically disabled users without driving permits are likely to benefit from FAVs. However, the potential for excessive distance errors will cause similar difficulties and threaten adoption.

In summary, we strongly believe that ONS will remain a main control mode for FAVs. However, ONS need to be significantly improved to address user experience issues. We hope that our findings will alert the automotive industry to pay attention to the latent “last-100-meters” issue, and provide effective and user-friendly solutions for future development.

Note

1. Using the <https://www.wjx.cn/> online survey service.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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